

Voice-Based AI Receptionist Agent for Smart College Communication

Using RAG and Agentic AI

Project Proposal — Phase 1 and Phase 2

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Abstract

Educational institutions receive numerous telephone inquiries every day regarding admissions, courses, departments, fee structure, placements, hostel facilities, academic schedules, and student services. Handling every call manually increases the workload of reception staff, leads to longer waiting times, and may result in inconsistent responses. To address these challenges, this project proposes a Voice-Based AI Receptionist Agent capable of answering calls autonomously using speech recognition, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), and Agentic AI.

In Phase 1, the voice agent operates on publicly available college information covering admissions, departments, faculty details, campus facilities, placements, events, and academic regulations. Incoming calls are handled via a SIP-based telephony stack (Asterisk 22.10.1 with MicroSIP and AudioSocket), which captures caller audio, converts speech to text using Faster-Whisper, retrieves relevant institutional knowledge from a ChromaDB vector store populated with 767 indexed chunks, generates a grounded response using a locally hosted Qwen3:4B language model via Ollama, and delivers the synthesised response back to the caller using Piper TTS. The agent is designed to answer only from the institutional knowledge base and to decline or seek clarification when the retrieved context is insufficient to support a confident response.

Phase 2 introduces an authenticated Agentic AI layer that integrates with the college ERP or DigiCampus platform to answer personalised student queries such as attendance, internal marks, fee status, examination results, and placement eligibility. It further implements intelligent call escalation to a human receptionist when a query requires administrative judgment or falls outside the agent's defined scope.

Phase 1 was evaluated on a set of 20 test queries, comprising 18 normal institutional information queries and 2 negative (out-of-scope) queries. The evaluation recorded a Hit@5 retrieval accuracy of 100%, an answer accuracy of 100%, and a refusal accuracy of 100% on out-of-scope queries. A separate latency evaluation measured an average end-to-end AI processing time of 21.35 seconds, comprising 0.25 seconds for speech-to-text, 0.07 seconds for RAG retrieval, 18.55 seconds for LLM generation, and 2.46 seconds for text-to-speech synthesis. LLM generation is the dominant contributor to latency and is identified as the primary bottleneck for future optimisation.

The proposed solution aims to reduce receptionist workload, provide consistent and knowledge-grounded responses, improve caller experience, and is designed to support continuous availability for routine inquiries when deployed as a continuously running service, while establishing a modular and evaluable AI pipeline suitable for extension toward personalised student services.

1. Background and Motivation

College reception desks are a critical point of contact for prospective students, parents, and current students, yet they operate under significant resource constraints. A single receptionist handles a mix of walk-in visitors, email inquiries, and telephone calls simultaneously, often under pressure during admissions season, examination periods, and placement drives. The result is extended hold times, inconsistent information delivery depending on which staff member answers, and complete unavailability outside office hours.

This project targets the telephone communication channel by introducing an intelligent voice agent that handles routine institutional queries autonomously. Phase 1 establishes the foundational voice-and-retrieval pipeline over a locally hosted architecture; Phase 2 extends it with authenticated, personalised student services and intelligent call escalation.

2. Problem Statement

College reception staff frequently handle repetitive telephone queries about institutional information such as fee structure, admission criteria, hostel availability, and department contacts. Manual handling of such queries can result in increased workload, repetitive tasks, longer waiting times, inconsistent information delivery depending on which staff member answers, and limited availability outside staffed hours.

Much of this information already exists on the college website and in institutional documents, but callers often require a conversational, telephone-based interface rather than a search interface. There is also no existing mechanism to connect telephone inquiries with institutional ERP data for personalised student queries.

This project therefore proposes an AI receptionist that combines telephony, speech recognition, RAG, and an LLM to answer public institutional queries (Phase 1), and later extends this capability using Agentic AI, authentication, institutional API integration, and human escalation (Phase 2).

Phase 1 Problem: Build a locally hosted voice AI that autonomously handles telephone inquiries about publicly available institutional information using Asterisk/SIP telephony, Faster-Whisper STT, a ChromaDB-backed RAG pipeline, a locally hosted Qwen3:4B LLM via Ollama, and Piper TTS — with no access to personal student data.

Phase 2 Problem: Extend the receptionist to handle authenticated personalised student queries using approved institutional APIs such as DigiCampus, and provide intelligent escalation to human reception staff when required.

3. Existing System

Typical college inquiry handling relies on human receptionists, static websites, manually accessed institutional documents, conventional telephone calls, and separate ERP or student portals for personalised information.

Limitations of this existing approach include:

- Repetitive receptionist workload for routine, repeatable queries.

- Dependency on staff availability, with no coverage outside office hours.
- Delays and inconsistent information delivery during periods of high call volume.
- Lack of a natural, voice-based automated interface for institutional information retrieval.
- Separation between publicly available information and personalised ERP/student services, with no unified conversational access point.

4. Proposed System

The proposed system is a two-phase Voice-Based AI Receptionist Agent.

- Phase 1: A knowledge-grounded public institutional voice receptionist that autonomously answers calls using RAG over public college information.
- Phase 2: An authenticated Agentic AI receptionist capable of personalised institutional services (via DigiCampus/API integration) and intelligent human escalation.

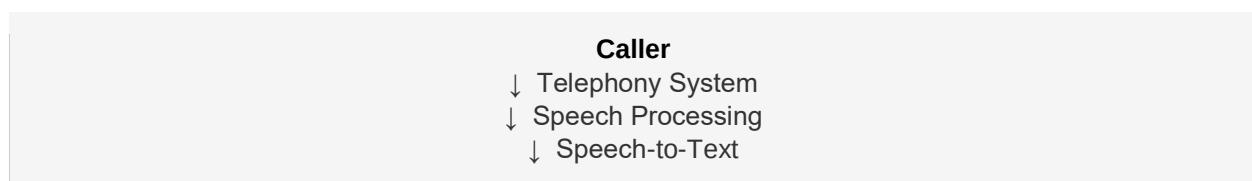
Phase 2 is designed as a modular extension of Phase 1 rather than a replacement, allowing the core voice pipeline (telephony, STT, TTS) to be reused while the decision and retrieval layers are extended with agentic tool-use capabilities.

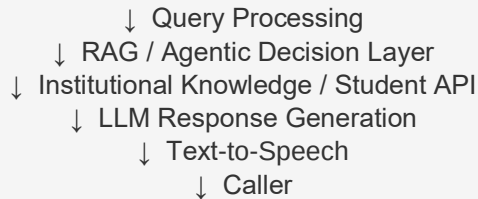
5. Objectives

- Automate handling of routine telephone inquiries directed at the college reception desk.
- Reduce receptionist workload for repeatable, publicly answerable queries.
- Provide consistent, knowledge-grounded institutional responses regardless of call volume.
- Enable natural-language voice interaction over standard SIP telephony infrastructure.
- Integrate speech processing with Retrieval-Augmented Generation and a locally hosted LLM.
- Develop a modular telephony AI architecture that supports independent upgrades of individual components.
- Evaluate retrieval accuracy, answer accuracy, and refusal behaviour on a defined test set.
- Study end-to-end AI processing latency across each pipeline stage.
- Extend the system toward authenticated, personalised student services in Phase 2.
- Introduce an Agentic AI decision layer in Phase 2 to route between public knowledge, personalised data, and human escalation.
- Enable future human receptionist escalation for queries the AI cannot safely or confidently resolve.

6. System Architecture

6.1 Conceptual Flow





6.2 Phase 1 Architecture (Implemented)

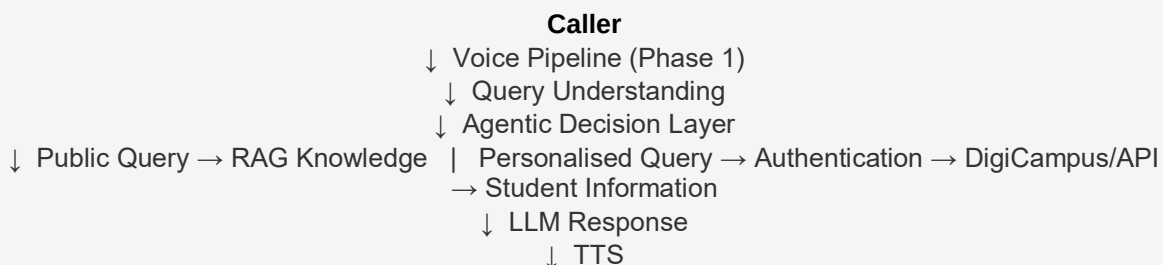
Phase 1 is implemented as a modular, locally hosted voice pipeline. The complete call path is:



Every major component has a distinct responsibility: Asterisk and MicroSIP handle SIP call signalling; AudioSocket transports raw PCM audio between Asterisk and the Python gateway; WebRTC VAD segments speech from silence; Faster-Whisper performs local, GPU-accelerated transcription; the RAG service embeds the query, retrieves and reranks relevant chunks from ChromaDB, and assembles grounded context; Ollama serves the Qwen3:4B model to generate the response; and Piper TTS synthesises the reply into telephony-compatible audio.

6.3 Phase 2 Architecture (Proposed)

Phase 2 extends the Phase 1 voice pipeline with a query-classification and Agentic Decision Layer that routes each caller query to public RAG knowledge, authenticated DigiCampus data, or human escalation:



↓ Caller
(If unresolved) → Human Receptionist Escalation

7. Phase 1 Workflow

1. The caller initiates a SIP call to the configured extension.
2. Asterisk 22.10.1 receives and answers the call from a MicroSIP softphone.
3. Asterisk establishes an AudioSocket connection with the Python audio gateway.
4. Incoming telephony audio is received by the gateway as raw PCM audio.
5. WebRTC VAD identifies speech segments within the audio stream.
6. Faster-Whisper (small model, CUDA-accelerated, float16 inference) transcribes the caller's speech into text.
7. The transcribed text query is passed to the Retrieval-Augmented Generation pipeline.
8. Sentence Transformer embeddings are used to retrieve semantically relevant institutional chunks from ChromaDB.
9. Retrieval results are reranked where necessary to improve relevance, with source and page metadata attached.
10. Retrieved contextual information, together with the query, is supplied to the locally hosted Qwen3:4B model through Ollama.
11. The LLM generates a response grounded in the retrieved institutional context. If sufficient information is unavailable, the agent is instructed to decline politely rather than speculate.
12. Piper TTS (en_US-lessac-medium) converts the generated response into speech.
13. The synthesised speech is converted into 8 kHz PCM telephony-compatible audio.
14. AudioSocket sends the response audio back to Asterisk.
15. Asterisk plays the response to the caller over the SIP call.

8. Phase 2 — Proposed Extension

Phase 2 is the planned extension of the Phase 1 system. It introduces Agentic AI so that the system can decide whether a query should be handled using public institutional RAG knowledge, authenticated student-specific institutional data, or human receptionist escalation.

8.1 Personalised Services

Subject to institutional approval and API access, Phase 2 may support authenticated retrieval of:

- Student attendance and subject-wise attendance
- Internal assessment marks
- Examination results
- Fee payment status and outstanding dues
- Placement eligibility
- Other institution-approved student information

These services depend on access to DigiCampus or another approved institutional API and are not part of the current Phase 1 scope.

8.2 Authentication (Conceptual)

Caller identity verification for Phase 2 may use institution-approved mechanisms such as Student ID, PIN, OTP, or DTMF keypad input, or another approved mechanism. The exact mechanism will depend on institutional approval and security requirements, and is not yet implemented.

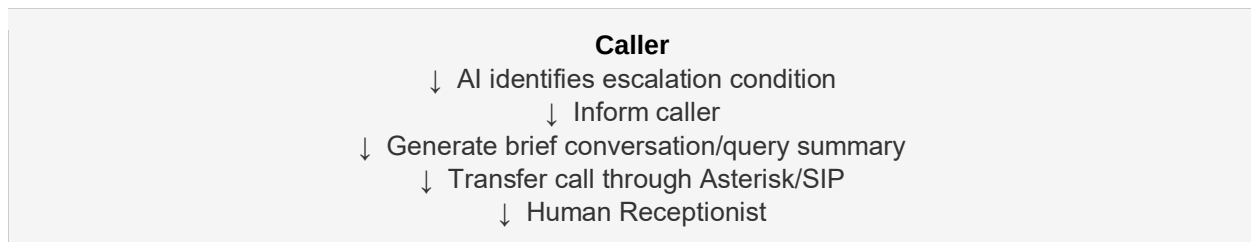
8.3 Agentic AI Decision Layer

The Agentic AI layer acts as a decision and tool-use layer rather than an unrestricted autonomous agent. Within predefined institutional tools, permissions, and policies, the agent may determine whether to:

- answer from the public RAG knowledge base
- request authentication
- access an approved student-data API
- request clarification
- refuse an unsupported operation
- escalate to a human receptionist

8.4 Human Receptionist Escalation (Planned)

Escalation to a human receptionist may occur when the caller explicitly requests a human, the query requires administrative approval or human judgement, the query is unsupported, authentication fails, requested data cannot be accessed, system confidence is insufficient, or a sensitive/exceptional situation arises.



This escalation flow is a planned Phase 2 capability and is not implemented in the current Phase 1 system.

9. Technology Stack

9.1 Phase 1 — Implemented Stack

Layer	Technology	Rationale
Telephony	Asterisk 22.10.1 + SIP + MicroSIP softphone	Open-source PBX providing a complete SIP-based local telephony environment for prototype development

Audio Transport	Asterisk AudioSocket	Native Asterisk protocol for raw PCM audio streaming to an external application over TCP
Voice Activity Detection	WebRTC VAD	Lightweight, real-time speech segment detection; avoids sending silence to the STT engine
Speech-to-Text	Faster-Whisper (small, CUDA, float16)	Local GPU-accelerated STT with strong English-language accuracy; avoids cloud API dependency
Web Scraping	Python + Requests + BeautifulSoup	Reliable extraction of institutional website content from static HTML pages
PDF Extraction	PyMuPDF + Tesseract OCR	Handles both text-native and scanned image-based PDF documents
Embeddings	Sentence Transformers	Open-source, locally hosted embedding model; no per-call embedding API cost
Vector Database	ChromaDB	Lightweight local vector store appropriate for single-institution corpus scale
LLM	Ollama + Qwen3:4B (locally hosted)	Fully local inference; no cloud API dependency; suitable for institutional data privacy in a prototype
Text-to-Speech	Piper TTS — en_US-lessac-medium	Fast, locally hosted neural TTS; produces natural speech output without external API calls
Backend	Python	Coordinates all pipeline components: audio gateway, VAD, STT, RAG, LLM, and TTS

9.2 Phase 2 — Planned Additions

Layer	Planned Technology	Purpose
Agentic Orchestration	LLM native function/tool calling or comparable agent framework	Enables the LLM to route between RAG retrieval and DigiCampus API tools dynamically
Caller Authentication	Student ID + PIN (DTMF) or OTP via SMS	Voice-channel identity verification before any personalised data is retrieved
DigiCampus Integration	Read-only REST API calls to DigiCampus endpoints	Live retrieval of attendance, marks, fee status, and placement eligibility per authenticated student
Call Transfer	Asterisk call transfer with pre-transfer transcript summary	Escalation to human receptionist with conversation context delivered before call connects
Audit Logging	Structured call logs in a local database	Enables accuracy monitoring, audit, and iterative improvement of the knowledge base

10. Knowledge Base and Data Sources

The Phase 1 knowledge base consists only of publicly available Rajalakshmi Engineering College information.

Website Content

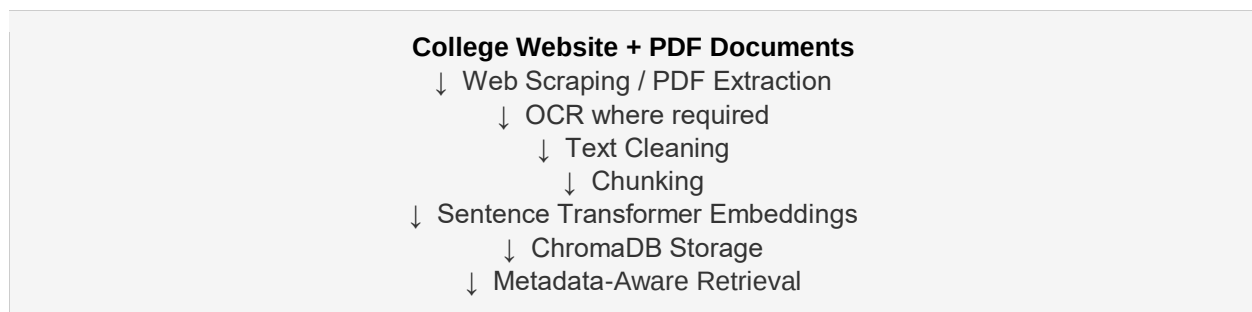
- Institutional pages covering: About, Academics, Admissions, Departments, Faculty, Placements, Research, Facilities, Events, and department-specific sub-pages, along with other relevant public institutional information.

PDF Documents

- Certificate for CGPA to Percentage Conversion
- Certificate for Medium of Instruction
- R2023 CSE Curriculum and Syllabus
- REC-2023 UG Regulations

Some source PDFs are scanned documents; Tesseract OCR is used when a PDF page does not contain machine-readable text. The Phase 1 knowledge base contains a total of 767 indexed chunks.

11. Knowledge Ingestion Pipeline



Website content is collected using Python with Requests and BeautifulSoup from publicly available institutional pages. PDF documents are parsed using PyMuPDF, with scanned or image-based PDFs processed through Tesseract OCR. Extracted text is cleaned to remove navigation artefacts and repeated headers, then chunked with overlap to preserve contextual continuity. Chunks are embedded using Sentence Transformers and stored in ChromaDB along with metadata such as source URL, document/file name, page number, and chunk information, which supports attribution and metadata-aware retrieval. A re-ingestion process can be run when the knowledge base requires updating.

12. RAG Architecture

The Retrieval-Augmented Generation system converts the caller's query into a vector representation, performs semantic similarity search in ChromaDB, retrieves relevant institutional chunks, and applies relevance-based reranking where necessary. The retrieved context is assembled and passed, along with the caller's question, to the Qwen3:4B model via Ollama, which generates an answer based primarily on the retrieved institutional information.

Grounding the LLM's response in retrieved institutional context reduces the likelihood of unsupported or fabricated answers compared to an ungrounded LLM. This does not eliminate the possibility of occasional errors, and the agent is instructed to decline or seek clarification when retrieved context is insufficient rather than speculate.

13. Telephony Architecture

Telephony is handled by Asterisk 22.10.1 acting as a local SIP PBX, with MicroSIP used as the softphone client for prototype calls. Asterisk routes inbound calls to a Python audio gateway over the AudioSocket protocol, which streams raw PCM audio bidirectionally between Asterisk and the gateway over TCP. This architecture keeps telephony signalling, audio transport, and AI processing as independently upgradeable modules.

14. Phase 2 Agentic Architecture

In Phase 2, an Agentic Decision Layer sits between query understanding and response generation. For each caller query, the layer determines whether the request can be answered from the public knowledge base, requires authenticated access to personalised student data via DigiCampus/API, or should be escalated to a human receptionist. This layer operates within predefined institutional tools, permissions, and policies rather than as an unrestricted autonomous agent.

15. Security and Privacy

15.1 Phase 1

- Uses only publicly available institutional information.
- Does not access personal student records.
- Does not require student authentication.

15.2 Phase 2

- Requires authenticated access before any personalised data is retrieved.
- Should follow least-privilege principles when accessing institutional APIs.
- Should use secure institutional APIs such as DigiCampus over authenticated, read-only channels.
- Should avoid exposing credentials or sensitive tokens through voice responses.
- Should handle sensitive student information securely and minimise retained data.
- Should maintain appropriate audit and logging controls.
- Requires institutional approval prior to implementation.

No claims are made here regarding formal regulatory compliance; specific compliance requirements would need to be assessed with institutional IT and legal stakeholders as part of Phase 2 planning.

16. Evaluation Methodology

Phase 1 was evaluated using a set of 20 test queries: 18 normal institutional information queries and 2 negative (out-of-scope) queries designed to test the agent's refusal behaviour. Retrieval quality was assessed using Hit@5, answer quality was assessed for the normal queries, and

refusal behaviour was assessed for the negative queries. A separate latency evaluation measured the average AI processing time contributed by each pipeline stage, from STT input through TTS output preparation.

17. Phase 1 Results

17.1 Retrieval and Answer Accuracy

Metric	Result
Total evaluation queries	20
Normal information queries	18
Negative / out-of-scope queries	2
Retrieval correct (Hit@5)	18 / 18 — 100%
Answer accuracy on normal queries	18 / 18 — 100% on the evaluated test set
Negative query handling (refusal accuracy)	2 / 2 — 100% on the evaluated test set

The system achieved 100% answer accuracy and retrieval Hit@5 on the defined evaluation dataset. These results reflect performance on this specific test set and should be interpreted accordingly; performance may vary across query types and phrasing not covered by the evaluation set.

17.2 Latency Evaluation

Pipeline Stage	Measured Average Latency
Speech-to-Text (Faster-Whisper)	0.25 seconds
RAG Retrieval	0.07 seconds
LLM Generation (Ollama + Qwen3:4B)	18.55 seconds
Text-to-Speech (Piper TTS)	2.46 seconds
Total AI Processing Latency	21.35 seconds

LLM generation is the primary latency bottleneck, contributing approximately 86.9% of the measured AI processing latency. These measurements were obtained in the local development environment and represent AI processing time from STT input through TTS audio preparation; they do not include telephony signalling or network transmission delays, and should not be read as the complete caller-perceived latency. Cloud GPU deployment and model or runtime optimisation are identified as potential approaches for improving latency in Phase 2.

18. Advantages

- Automated responses to routine institutional queries reduce dependency on manual reception staff.
- Reduced repetitive workload for reception staff during high call-volume periods.

- Consistent information delivery, since the agent answers only from indexed institutional content.
- A natural voice interface for institutional information retrieval.
- Knowledge-grounded LLM responses via Retrieval-Augmented Generation.
- Modular architecture allowing individual components (STT, LLM, TTS, retrieval) to be upgraded independently.
- Local, open-source AI components in Phase 1 reduce dependency on paid cloud AI APIs during development.
- Extensibility toward Phase 2 personalised student services without redesigning the core pipeline.
- Planned future human escalation for cases the AI cannot safely resolve.
- Measurable evaluation: retrieval accuracy, answer accuracy, refusal behaviour, and latency have been quantified.

These advantages are stated with academic caution: the system does not claim zero hallucination, perfect reliability, guaranteed 24×7 availability, unlimited concurrent call handling, or production readiness at this stage.

19. Limitations and Risks

19.1 Phase 1 Limitations

- Local hardware dependency: Phase 1 inference performance is bounded by the available local GPU, and LLM generation latency (18.55 seconds average) is not suitable for conversationally responsive deployment on the current hardware configuration.
- STT accuracy may vary with accents, background noise, and microphone/telephony audio quality.
- Response quality depends on the completeness and freshness of the ingested institutional data.
- Phase 1 does not access personalised student information; such queries cannot be answered until Phase 2 is implemented.
- The current deployment is a prototype telephony environment rather than a production system.

19.2 Phase 2 Limitations and Dependencies

- Availability of a documented DigiCampus/API interface for student data.
- Institutional approval for API access and authentication design.
- Design and approval of a secure caller authentication mechanism.
- Student-data privacy requirements and secure handling of sensitive information.
- Call transfer configuration to a defined human-receptionist endpoint.
- Production deployment infrastructure beyond the current prototype environment.
- Secure logging and audit requirements for Phase 2 operation.

20. Phased Scope and Milestones

Phase 1 implementation and evaluation have been completed. Mentor review and approval constitute the transition milestone before Phase 2 planning begins.

Milestone	Description	Status
M1	Website and selected PDF ingestion pipeline completed; knowledge base of 767 chunks populated and searchable in ChromaDB.	COMPLETED
M2	RAG backend integrated with locally hosted Qwen3:4B LLM via Ollama; evaluated using text-based test queries.	COMPLETED
M3	Faster-Whisper → RAG → LLM → Piper TTS voice pipeline integrated end-to-end.	COMPLETED
M4	Asterisk/SIP/MicroSIP/AudioSocket telephony integration completed; end-to-end call flow operational.	COMPLETED
M5	Retrieval, answer accuracy, refusal, and latency evaluation completed.	COMPLETED
M6	Mentor review and approval of Phase 1 results; decision on Phase 2 scope and DigiCampus API access.	PENDING
M7	Caller authentication design and implementation.	PLANNED
M8	DigiCampus/API integration for approved personalised queries.	PLANNED
M9	Agentic routing and human receptionist escalation flow.	PLANNED
M10	Complete Phase 2 evaluation, final documentation, and demonstration.	PLANNED

21. SDG Alignment

- SDG 04 — Quality Education: The system improves accessibility to institutional and academic information for students, parents, and prospective applicants.
- SDG 09 — Industry, Innovation and Infrastructure: The project applies AI, telephony, and digital infrastructure to institutional communication.
- SDG 16 — Peace, Justice and Strong Institutions: The system supports consistent, transparent, and accessible institutional information delivery.

22. Expected Outcomes

The expected outcome of Phase 2 is a unified AI receptionist capable of answering public institutional questions, securely identifying authorised students, retrieving approved personalised information, deciding which institutional tool or data source to use for a given query, and escalating cases that require human attention. These capabilities are proposed extensions and are not current Phase 1 achievements.

23. Mentor Discussion / Phase 2 Dependencies

16. DigiCampus API access: Does the college IT department expose a documented REST API for student data, or would Phase 2 require an alternative approach such as a read-only database connector or an intermediary microservice?

17. Authentication mechanism: What caller identity verification mechanism is institutionally acceptable for a voice channel (Student ID + DTMF PIN, OTP via SMS, or another approach)?
18. Human receptionist transfer endpoint: Does the college telephony infrastructure support SIP-based call transfer to a defined reception extension?
19. Data retention and privacy: Should call transcripts and anonymised query logs be stored for evaluation and quality-improvement purposes, and if so, under what retention policy?
20. Phase 2 feature priority: Of the planned capabilities (authenticated student queries, personalised data retrieval, human escalation), which should be prioritised first given API access constraints and development time?
21. Latency optimisation approach: Should Phase 2 target cloud GPU deployment for LLM inference, or should the system remain locally hosted with model-level optimisation (quantisation, smaller model variants)?
22. Phase 2 evaluation criteria: What metrics and acceptance thresholds should be used to evaluate Phase 2 (e.g., authentication success rate, DigiCampus query accuracy, escalation correctness)?

24. Academic Value

This project spans multiple technical disciplines that are rarely combined in a single undergraduate project.

Skill Domain	Phase 1 — Demonstrated	Phase 2 — Planned
RAG Engineering	Chunking, Sentence Transformer embeddings, ChromaDB retrieval, relevance reranking, source metadata, grounding constraints	—
Document Ingestion	Web scraping (BeautifulSoup), PDF parsing (PyMuPDF), OCR (Tesseract), content cleaning	—
Speech Pipeline	Real-time STT (Faster-Whisper), TTS (Piper), WebRTC VAD, 8 kHz PCM audio handling	—
Telephony	Asterisk 22.10.1, SIP, MicroSIP, AudioSocket, audio gateway implementation	Call transfer, escalation logic
LLM Integration	Locally hosted LLM via Ollama (Qwen3:4B), prompt grounding, refusal handling	Tool calling, agentic routing
Evaluation Methodology	Retrieval accuracy, answer accuracy, refusal accuracy, latency measurement per component	Phase 2 evaluation design
Security and Privacy	Knowledge base restricted to public data; no student PII processed	Authentication, per-user data isolation, audit logging

Agentic AI	—	Tool-use orchestration, DigiCampus API integration, confidence-based routing
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The combination of real telephony integration (Asterisk/SIP), a fully local AI stack (STT, retrieval, LLM, TTS), and a quantified evaluation methodology — applied to a genuine institutional problem — provides a technically credible and independently presentable project at the undergraduate final-year level.

25. Conclusion

Phase 1 of the Voice-Based AI Receptionist Agent has been implemented and evaluated. The system establishes a complete, locally hosted voice pipeline capable of receiving SIP calls, understanding natural-language queries, retrieving relevant institutional knowledge, generating grounded responses, and delivering them as synthesised speech, all without accessing personal student data.

The Phase 1 evaluation on a 20-query test set recorded 100% retrieval accuracy (Hit@5), 100% answer accuracy, and 100% refusal accuracy on out-of-scope queries. The measured average AI processing latency is 21.35 seconds, with LLM generation identified as the primary bottleneck. Latency reduction through cloud GPU deployment or model optimisation is a defined objective for Phase 2.

Phase 2, planned following mentor review and institutional approval, will introduce authenticated student data retrieval via DigiCampus API integration, Agentic AI tool-calling, and intelligent call escalation to a human receptionist. Phase 1 approval and a decision on Phase 2 scope remain the immediate next steps.